Power for One-Proportion Test

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Introduction

Power analysis for the one-proportion test computes the sample size required to detect a departure from a null proportion p_0 with adequate probability. The one-proportion test arises constantly in clinical research — testing whether a single-arm response rate exceeds a historical control benchmark, whether a quality-control defect rate departs from a tolerance threshold, whether a survey-derived proportion differs from a reference. Two complementary approaches are widely used: the Normal-approximation power formula, which gives clean closed-form expressions in terms of Cohen's h effect size, and exact-binomial power, which enumerates the binomial distribution and is preferred for small samples or extreme proportions where the Normal approximation is unreliable.

Prerequisites

A working understanding of the one-proportion test, the binomial distribution, the Normal approximation to the binomial, and Cohen's h as the standardised effect-size measure for proportion tests.

Theory

For the test of $H_0 : p = p_0$ versus $H_1 : p = p_1$, Cohen's standardised proportion-difference effect size is

$$h = 2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_0},$$

which arc-sine transforms each proportion onto a scale where its variance is approximately constant. The Normal-approximation sample-size formula at two-sided α and power $1 - \beta$ is

$$n \approx \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{h} \right)^2 .$$

Cohen's conventional benchmarks for h are 0.20 (small), 0.50 (medium), and 0.80 (large). Exact binomial power evaluates the binomial CDF under p_1 and sums probabilities in the rejection region defined by the binomial CDF under p_0.

Assumptions

Trials are independent Bernoulli with constant success probability; for the Normal-approximation formula, $np_0 \geq 10$ and $n(1 - p_0) \geq 10$ to ensure the approximation is adequate; for exact binomial power these conditions are not needed.

R Implementation

```
library(pwr)

h <- ES.h(p1 = 0.40, p2 = 0.25)
pwr.p.test(h = h, sig.level = 0.05, power = 0.80, alternative = "two.sided")
```

```
exact_binom_power <- function(n, p0, p1, alpha = 0.05) {
  k_reject <- sum(dbinom(0:n, n, p0) <= alpha)
  crit_lo <- qbinom(alpha/2, n, p0)
  crit_hi <- qbinom(1 - alpha/2, n, p0)
  power_lo <- pbinom(crit_lo - 1, n, p1)
  power_hi <- 1 - pbinom(crit_hi, n, p1)
  power_lo + power_hi
}
exact_binom_power(n = 80, p0 = 0.25, p1 = 0.40)
```

Output & Results

`pwr.p.test()` returns the sample size required under the Normal-approximation formula; the custom exact-binomial function then verifies the calculation by direct enumeration. For the example, the Normal approximation gives $n = 82$ and the exact calculation confirms 80 % power at $n = 80$ — close agreement that is typical when the Normal-approximation conditions are satisfied.

Interpretation

A reporting sentence: "To detect a response rate of 40 % against a historical-control null of 25 % with 80 % power at two-sided $\alpha = 0.05$, $n = 82$ participants are required (Normal-approximation calculation, Cohen's $h = 0.32$). Exact binomial power at $n = 80$ is 80 %, confirming the Normal-approximation result. The protocol enrols 90 to allow for 10 % attrition. Sensitivity analyses across $p_1 \in [0.35, 0.45]$ are reported in the supplement." Always report sensitivity over plausible p_1.

Practical Tips

- For small n (typically < 30) or extreme proportions (very close to 0 or 1), prefer exact binomial power calculations to the Normal approximation; the approximation is unreliable in these regimes and can give misleading sample-size recommendations.
- Cohen's h ranges in $[0, \pi]$ with conventional benchmarks 0.20 (small), 0.50 (medium), and 0.80 (large); these are guidelines for translating effect-size language into sample-size calculations and should be tied to substantive clinical or scientific meaning.
- One-sided tests require less sample size than two-sided tests if the direction of effect is pre-specified and scientifically justified; the trade-off is that a result in the unexpected direction cannot be reported as significant.
- Sensitivity analysis over a range of plausible p_1 values is standard; protocols with a single point estimate of p_1 are increasingly flagged by reviewers, who expect a defensible bracket and a justification of the lower bound used for the sample-size determination.
- For p_0 near 0.5, the test is approximately symmetric and the variance is maximised, so power per unit of n is at its maximum; for p_0 near 0 or 1 the variance is small and the test can be very efficient at detecting small absolute differences.
- Continuity-correction options (Yates correction) marginally affect Normal-approximation power calculations; modern software defaults vary, so check what your sample-size tool is computing.

R Packages Used

`pwr::pwr.p.test()` and `pwr::ES.h()` for canonical Cohen's h Normal-approximation calculations; `binom::power.binom()` for exact binomial sample-size and power; `Hmisc::bpower()` for an alternative implementation; `MESS::power_prop_test()` for fast exact and approximate calculations; `Superpower` and `simr` for simulation-based extensions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation